

Satellite communication used in the non-geostationary Fixed-satellite service systems and Earth-exploration satellite service (passive) systems

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Abstract—This paper is about the satellite communication and focused on compatibility study between non-geostationary fixed-satellite service systems and Earth-exploration satellite service (passive) systems. It mainly carried out the analysis of the nearby band interferences to geostationary passive sensors in the Earth-exploration satellite service in the 50.2-50.4 GHz frequency band that may occur from the fixed-satellite service systems in the 47.2-50.2GHz and 50.4-51.4 GHz frequency band in low Earth orbit and medium Earth orbit respectively. The research made the conclusion that the avoidance angle as a possible interference mitigation technique could be used between the non-geostationary fixed-satellite service systems and the geostationary Earth-exploration satellite service systems. The result shows that, in order to achieve the compatibility in the same frequency band, the geostationary avoidance angle for the non-geostationary fixed-satellite service earth stations should not be less than 10°.

Keywords—satellite communication, compatibility study, non-geostationary fixed-satellite service, Earth-exploration satellite service, interference mitigation technique

I. INTRODUCTION

With the development of technology, more and more satellite systems provide services for human life. More and more satellite communications are used in smart transportation and logistics. The satellite meteorological service serves a wide range of maritime transport, especially transoceanic transport.

International Telecommunication Union (ITU) is the United Nations specialized agency for information and communication technologies (ICTs), allocating global radio spectrum and satellite orbits, developing the technical standards that ensure networks and technologies seamlessly interconnect, and striving to improve access to ICTs to underserved communities worldwide. The ITU Radiocommunication Bureau (BR) as the implementing agency of the ITU Radiocommunication Sector (ITU-R) responses for coordinating radiocommunication services and managing radio spectrum and satellite orbit resources at the international level. It manages and maintains the order of international satellite frequency orbit resources in accordance with the Radio Regulations and the other laws.

Currently in the Radio Regulations (RR)[1], there are twenty-one space radio services were defined, including fixed-satellite service (FSS), mobile-satellite service (MSS), broadcasting-satellite service (BSS), Earth exploration-satellite service (EESS) and so on. Among them, EESS is defined as a radiocommunication service between earth stations and one or more space stations, and a passive sensor on Earth satellite which obtains information relating to the characteristics of the Earth and its natural phenomena, including data relating to the state of the environment, is one of such EESS links. As an important space power, China has developed many satellite projects rapidly in recent years, and China became the No.1 of the world in the number of space launches in 2018. More than half of the recent launches were scientific-related services satellites, such as Dark Matter Particle Explorer (DAMPE) and FENGYUN series satellites, in most of which the EESS passive sensors were deployed.

The application of multiple space services has also led to an increase in the use of space radio frequencies. For example, the 50.2-50.4 GHz frequency band is allocated worldwide to the EESS (passive) on a primary basis in the Table of Frequency Allocations of RR Article 5 while the FSS allocations in the bands 47.2-50.2 GHz (Earth-to-space) and 50.4-51.4 GHz (Earth-to-space) are adjacent to the EESS (passive) band. The frequency band 50.2-50.4 GHz is critical for passive remote sensing. This EESS (passive) band is used worldwide for the measurement of atmospheric temperature profiles and cloud detection. It is also used as a calibration band for numerous of other EESS (passive) bands. More crucially, this band is used for collecting data critical to Numerical Weather Models which are utilized for the prediction of hurricane trajectories and probability of tornado development. At the same time, non-geostationary (non-GSO) FSS develops rapidly, especially the Low Earth orbit (LEO) FSS constellation. Many countries have plans to deploy next-generation broadband constellation systems in the 50 GHz band for the use of broadband satellite Internet. Therefore, the impact on the adjacent frequency band used by the EESS (passive) is more severe.

This paper focused on compatibility study between non-GSO FSS systems and EESS (passive) systems. The study examined the potential interference from non-GSO FSS systems in order to derivate an appropriate out of band

emission limits to protect the EESS (passive) in the 50.2-50.4 GHz band. The study also indicated interference mitigation techniques and derived conditions for sharing between the two systems.

II. COMPATIBILITY CALCULATION

A. Method

This study provides an analysis of the impact of unwanted emissions from non-GSO FSS systems into geostationary (GSO) EESS (passive) systems operating in the band 50.2-50.4 GHz.

The study looks at the interference statistics from two non-GSO systems, LEO type and medium Earth orbit (MEO) type into a GSO passive sensor. The following scenarios were analyzed:

- Interference from an equatorial MEO satellite system into a receiving GSO passive sensor.
- Interference from a LEO satellite system into a receiving GSO passive sensor.

The interference power, P , from one FSS earth station received by the GSO passive sensor as follow:

$$P = \frac{P_t G_t(\varphi_t)}{4\pi R^2 L_a} A = \frac{P_t G_t(\varphi_t)}{4\pi R^2 L_a} \frac{\lambda^2 G_r(\varphi_r)}{4\pi} \quad (1)$$

where

P_t : transmitted power of the FSS earth station in the 50.2-50.4 GHz band;

G_t : antenna gain pattern of the FSS earth station;

φ_t : the off-axis angle of the GSO passive sensor from the FSS earth station antenna boresight;

R : distance between the FSS earth station and the GSO passive sensor;

L_a : atmosphere attenuation loss;

A : effective antenna aperture of the GSO passive sensor;

λ : wavelength;

G_r : antenna gain pattern of the GSO passive sensor;

φ_r : the off-axis angle of the FSS earth station from the GSO passive sensor antenna boresight.

In the above equation, the values of G_t , G_r and R will change with the relative positions among the FSS earth station, the FSS satellite and the GSO passive sensor. Therefore these variables must be determined firstly. Since the locations of the FSS earth station, the FSS satellite, the GSO passive sensor and its observed surface point in the geodetic coordinate system are known at any moment, the geodetic location (i.e. latitude, longitude and height) of each point can be converted to the location (i.e. x , y and z) in geocentric Cartesian coordinate system. Then the distance between any two points can be calculated, and the two off-axis angles can be obtained by using the Law of Cosines.

So the aggregated interference power, P_{ag} , from all of the FSS earth stations that can be received by the GSO passive sensor is calculated as:

$$P_{ag} = 10 \lg(\sum P_i) \quad (2)$$

where

P_i : interference power from the i^{th} FSS earth station.

B. EESS sensor characteristics

ITU-R Recommendation RS.1861[4] only provides the technical and operational characteristics of non-GSO EESS (passive) systems. Besides non-GSO passive sensors, some countries are considering the development of GSO radiometers which will cover 50.2-50.4 GHz band. Table I lists the technical and operational characteristics of such two GSO EESS (passive) systems.

TABLE I. CHARACTERISTICS OF GSO EESS (PASSIVE) SENSORS IN THE 50.2-50.4 GHz BAND

Mission	Sensor A	Sensor B
Orbital position	GSO satellite, longitude to be shifted during the satellite lifetime	GSO satellite, longitude to be shifted during the satellite lifetime
Channels center and frequency bandwidth	50.3 GHz (200 MHz bandwidth)	50.3 GHz (180 MHz bandwidth)
Antenna type	2D synthetic aperture	Parabolic reflector
Effective antenna size	5 m	3 m
Maximum beam gain	66 dBi	
Polarization	H	V+H
-3 dB beamwidth	0.1° IFOV: 50 km at sub-satellite point	IFOV: 81 km at sub-satellite point
Period of full-disk scan mode	2D electronic scan in 10 min to form a full disk coverage from the GSO position	Each 15 min a sector of 1/12 of the Earth's disk will be imaged, the sector being movable within the disk

In this study, only GSO sensor A was considered. Sensor A is a two dimension synthetic aperture microwave radiometer, and it can point to any surface location on the earth disk very quickly through electronic scanning. The relationship between the received antenna gains of sensor A and the off-axis angles is shown in Fig. 1.

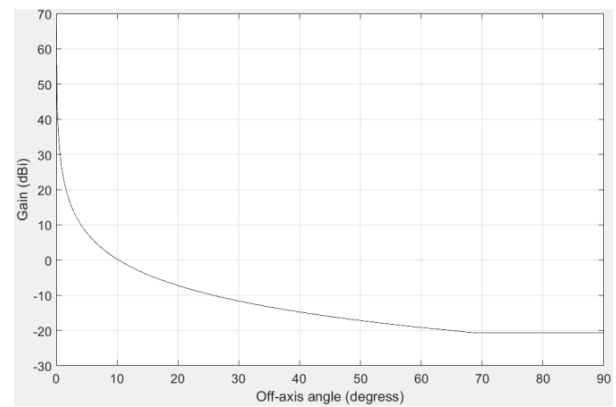


Fig. 1. The antenna gain pattern of the GSO passive sensor A

The protection criterion for EESS (passive) systems in the 50.2-50.4 GHz band is defined that a received power spectral density of -166 dBW/200 MHz is exceeded less than

0.01% of the time where the sensor footprint within a measurement area of 2 000 000 km², as indicated in ITU-R Recommendation RS.2017[5]. In the following simulation, the time ratio or the data availability is calculated as the percentage of the number of observations of sensor A which received aggregated interference power are not less than the above power spectral density related to the total number of observations.

C. Non-GSO FSS orbital and system characteristics

As noted above, two different constellations at different altitudes were simulated to determine their impact on EESS (passive) operations in the 50.2-50.4 GHz band. These constellations are representative of filings to the ITU: 1) a LEO at 1200 km; 2) a MEO at 8,062 km. Table II contains the orbital parameters of the constellations under this study.

The technical characteristics of the two constellations earth stations are given in Table III. The relationships between the off-axis angles and the transmitted antenna gains of LEO and MEO earth stations are shown in Fig. 2. The FSS earth station maximum antenna gain is assumed to 56.9 dBi. The unwanted emission power considered is -20 dBW/200 MHz, in line to the one given in Resolution 750 for antenna gains less than 57 dBi.

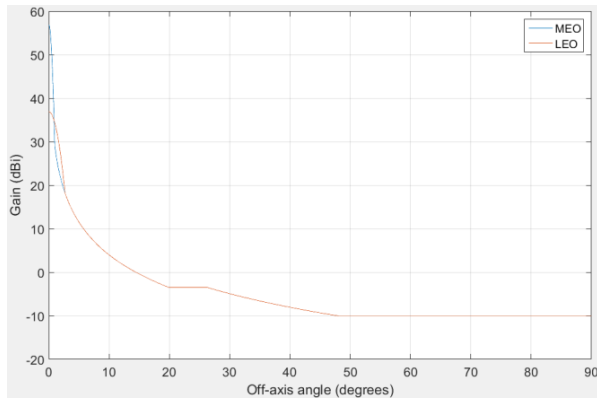


Fig. 2. The antenna gain patterns of the non-GSO FSS earth stations

Furthermore, three types tracking strategy of the above-mentioned non-GSO system have been considered:

1) Type 1:

- Minimum elevation Angle: 20°
- GSO avoidance angle: 0°
- The related satellite is chosen based on the highest elevation angle.

2) Type 2:

- Minimum elevation Angle: 20°
- GSO avoidance angle: 2°
- The related satellite is chosen based on the highest elevation angle.

3) Type 3:

- Minimum elevation Angle: 45°
- GSO avoidance angle: 10°
- The related satellite is chosen based on the highest elevation angle.

TABLE II. ORBITAL CHARACTERISTICS OF FSS NON-GSO SYSTEMS

Constellation	#1	#2
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Inclination (degree)	87.9	0
Altitude (km)	1200	8062
Number of Planes	18	1
Number of Satellites/plane	40	24
Total number of satellites	720	24
Number of beams per satellite	8	8

TABLE III. NON-GSO SYSTEM PARAMETERS FOR EARTH STATIONS

	MEO earth station	LEO earth station
Antenna size (m)	1.8	0.2
Antenna gain (dBi)	56.9	36.9
Gain pattern	ITU-R S.465-6[2]	ITU-R S.465-6[2]

D. Analysis for the MEO FSS constellation

For the MEO FSS constellation, 91 earth stations in total were deployed with a bias towards the equator (red stars as shown in Fig.3). The considered measurement area is located at (0°N, 25°E), and the GSO EESS satellite deployed sensor A is also located at this point. Fig.4 provides the measurement area (in blue), and location of the earth stations around the area.

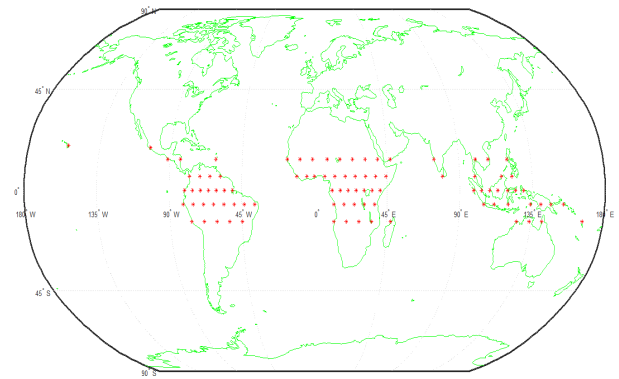


Fig. 3. Deployment of MEO FSS earth stations

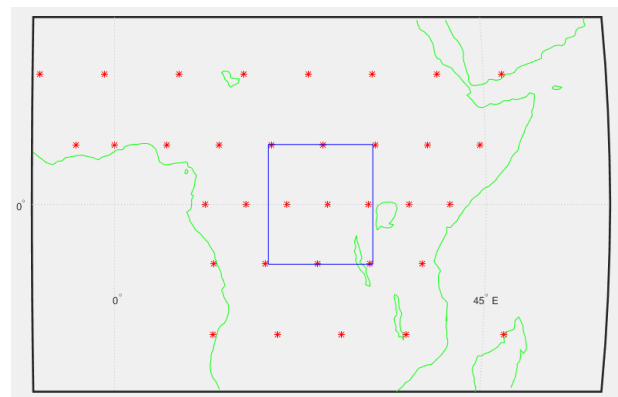


Fig. 4. Location of measurement area for MEO FSS

The MEO constellation orbits were simulated over 6 hours with a time step of 0.5 seconds. Atmospheric attenuations were also considered in the simulation, using the simplified model in ITU-R Recommendation P.676-11[3]. The cumulative distribution functions (CDF) obtained for different types of tracking strategy are provided in Fig. 5.

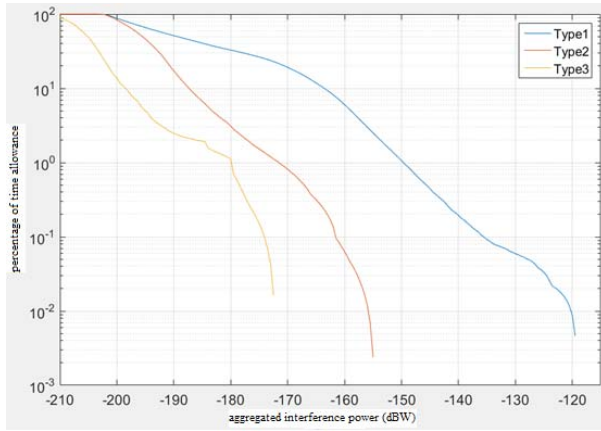


Fig. 5. CDF of simulation results for MEO FSS

The worst-case results are obtained when the type 1 tracking strategy is used. The protection criterion for EESS (passive) systems is exceeded by 46 dB for 0.01% of the time in this case. When the GSO avoidance angle increased to 2° , the protection criterion is exceeded by 10 dB. And the protection criterion is exceeded by -6 dB, when the GSO avoidance angle increased to 10° .

E. Analysis for the LEO FSS constellation

For the LEO FSS constellation, it is assumed that each satellite possesses 8 steerable beams, and that each beam just communicates with 1 earth station. As a result, in total 5600 earth stations were deployed for the analysis (as shown in Fig.6). The considered measurement area and the GSO EESS satellite are both located at $(0^\circ\text{N}, 25^\circ\text{E})$. Fig.7 provides the measurement area (in blue), and location of the earth stations around the area.

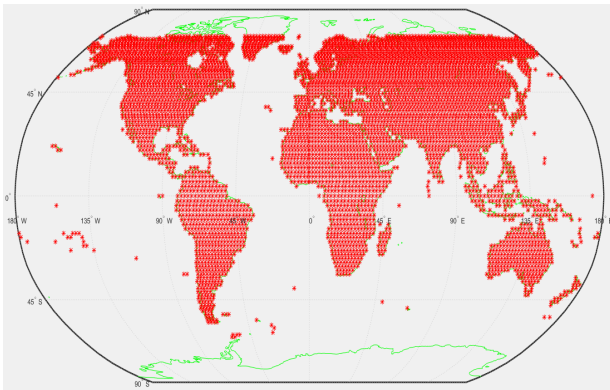


Fig. 6. Deployment of LEO FSS earth stations

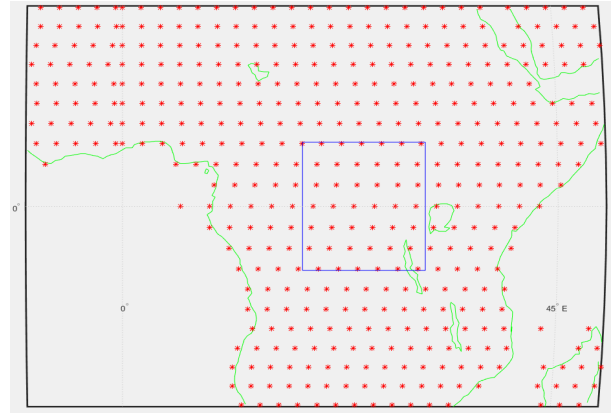


Fig. 7. Location of measurement area for LEO FSS

The LEO constellation orbits were simulated over 1 day with a time step of 0.5 seconds. Atmospheric attenuations were also considered in the simulation, using the simplified model in ITU-R Recommendation P.676-11[3]. The cumulative distribution functions obtained for different types of tracking strategy are provided in Fig. 8.

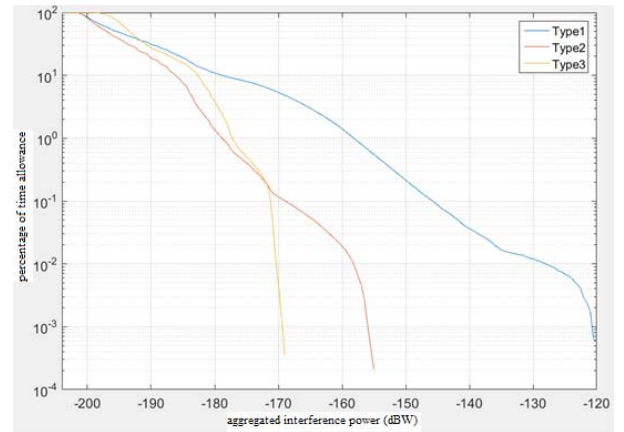


Fig. 8. CDF of simulation results for LEO FSS

The worst-case results are obtained when the type 1 tracking strategy is used. The protection criterion for EESS (passive) systems is exceeded by 38 dB for 0.01% of the time in this case. When the GSO avoidance angle increased to 2° , the protection criterion is exceeded by 8 dB. And the protection criterion is exceeded by -4.5 dB, when the GSO avoidance angle increased to 10° .

III. CONCLUSION

Based on the above analysis for the LEO and MEO FSS constellations, non-GSO FSS systems might exceed the protection criteria to the EESS (passive) operating in the 50.2-50.4 GHz band.

According to the results of the compatibility study, applying interference avoidance angle can reduce the interference of non-GSO FSS system to EESS (passive) systems. The maximum interference occurred when the interference avoidance angle is 0. As the interference avoidance angle increases, the interference of non-GSO FSS system to EESS (passive) systems was decreased.

The results of this study have shown that, in order to protect GSO EESS (passive) systems, the GSO avoidance angle for the non-GSO FSS earth stations should not be less

than 10° when the out of band emissions from each FSS earth station is limited to -20 dBW/200 MHz.

REFERENCES

- [1] International Telecommunication Union, "Radio Regulations", 2016.
- [2] International Telecommunication Union, Recommendation ITU-R S.465-6, "Reference radiation pattern for earth station antennas in the fixed-satellite service for use in coordination and interference assessment in the frequency range from 2 to 31 GHz", 6th ed., Geneva, 2010.
- [3] International Telecommunication Union, Recommendation ITU-R P.676-11, "Attenuation by atmospheric gases", 11th ed., Geneva, 2016.
- [4] International Telecommunication Union, Recommendation ITU-R RS.1861, "Typical technical and operational characteristics of Earth exploration-satellite service (passive) systems using allocations between 1.4 and 275 GHz", 1st ed., Geneva, 2010.
- [5] International Telecommunication Union, Recommendation ITU-R RS.2017, "Performance and interference criteria for satellite passive remote sensing", 1st ed., Geneva, 2012.